

Lyapunov Stability and Stability Regions of Nonlinear Dynamical Systems

3.1 INTRODUCTION

Stability is a fundamental subject that unifies engineering and the sciences. This subject has been regarded by many as a fascinating and difficult problem of human culture. As such, there are at least 50 different terms for stability concepts used in the literature. Stability is a very broad subject, and the concept of stability can be formulated in a variety of ways depending on the intended use of stability analysis and design (Alberto and Chiang, 2007; DeCarlo et al., 2000; Hahn, 1967; La Salle and Lefschetz, 1961; May, 1973; Michel et al., 1982). In this chapter, we review some relevant stability concepts from the nonlinear dynamical systems theory. Some of these concepts and their implications will be discussed in detail.

Another important subject related to stability is the stability region of nonlinear dynamical systems. Indeed, the problem of determining stability regions (regions of attraction) of nonlinear dynamical systems is of fundamental importance for many disciplines in engineering and in the sciences (Athay et al., 1984; Chiang et al., 1988; Genesio et al., 1985; Loparo and Blankenship, 1978; Saha et al., 1997; Sastry, 1999; Vu and Liu, 1992). For instance, knowledge of the stability region is essential in the development of direct methods for power system transient stability analysis. Indeed, this problem is at the heart of direct methods. A comprehensive theory of stability regions for the (autonomous) dynamical system will be derived in this chapter. Most of the proofs presented in this chapter are taken from Chiang et al. (1987, 1988) and from Chiang and Thorp (1989b). The topic of estimating stability regions for higher-dimension nonlinear dynamical systems will be discussed in Chapter 5.

3.2 EQUILIBRIUM POINTS AND LYAPUNOV STABILITY

We consider the following (autonomous) nonlinear dynamical system:

$$\dot{x} = f(x), x \in \mathfrak{R}^n. \quad (3.1)$$

It is natural to assume the function (i.e., the vector field) $f: \mathfrak{R}^n \rightarrow \mathfrak{R}^n$ satisfies a sufficient condition for the existence and uniqueness of a solution. The solution curve of Equation 3.1 starting from x at $t = 0$ is called the system trajectory starting from x and is denoted by $\phi(x, \cdot)$. The system trajectory starting from x is a function of time; given a specified time, the system trajectory function maps the specified time into a point in the state space. $\bar{x} \in \mathfrak{R}^n$ is said to be an *equilibrium solution* of Equation 3.1 if $f(\bar{x}) = 0$; that is, the equilibrium point is a solution that does not change in time. Hence, equilibrium points are degenerated solution curves that do not move.

The stability property of an equilibrium point will be discussed next. We first review the concepts of (Lyapunov) stability and asymptotic stability.

Definition: Lyapunov Stability

An equilibrium point $\bar{x} \in \mathfrak{R}^n$ of Equation 3.1 is said to be (*Lyapunov*) *stable* if, for each open neighborhood U of $\bar{x} \in \mathfrak{R}^n$, there exists an open neighborhood V of $\bar{x} \in \mathfrak{R}^n$ such that, for all $x \in V$ and for all $t > 0$, $\phi_t(x) \in U$. Equivalently, $\phi_t(x) \in U$ for all $t > 0$. Otherwise, $\bar{x}$ is unstable.

The concept of (Lyapunov) stability is illustrated in Figure 3.1. Intuitively speaking, an equilibrium point is stable if nearby trajectories stay nearby. In many applications, the requirement of “nearby trajectories stay nearby” is not sufficient; instead, the requirement becomes “nearby trajectories stay nearby and all converge to the equilibrium point.” Under this situation, the concept of (Lyapunov) stability can be sharpened into the concept of asymptotic stability as defined in the following.

Definition: Asymptotic Stability

An equilibrium point $\bar{x} \in \mathfrak{R}^n$ of Equation 3.1 is said to be *asymptotically stable* if there exists an open neighborhood U of $\bar{x} \in \mathfrak{R}^n$, such that, for all $x \in U$ and for all

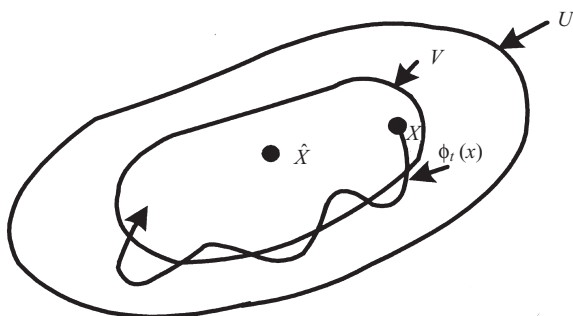


Figure 3.1 An illustration of the definition of (Lyapunov) stability.

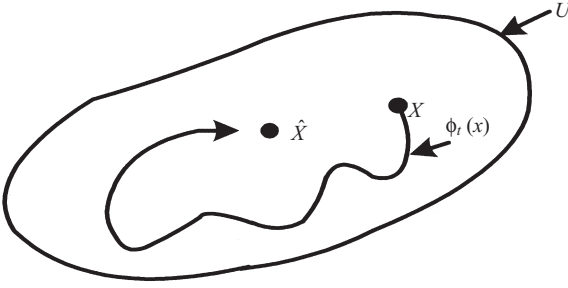


Figure 3.2 An illustration of the definition of asymptotic stability.

$t > 0$, $\phi_t(x) \in U$. In addition, every trajectory $\phi_t(x)$ starting from this neighborhood converges to the equilibrium point $\bar{x}$.

Equivalently,

- (i) $\phi_t(x) \in U$, $t > 0$ and
- (ii) $\lim_{t \rightarrow \infty} \|\phi_t(x) - \bar{x}\| = 0$.

The concept of asymptotic stability is illustrated in Figure 3.2. Intuitively speaking, an equilibrium point is asymptotically stable if it is the sink of nearby trajectories. An asymptotically stable equilibrium point is also termed a sink in the classical literature of nonlinear dynamical systems (Athay et al., 1984; Guckenheimer and Holmes, 1983; Hirsch and Smale, 1974).

In order to determine the stability of $\bar{x}(t)$, we must understand the nature of solutions near $\bar{x}(t)$. Let

$$x(t) = \bar{x}(t) + y(t). \quad (3.2)$$

Substituting Equation 3.2 into Equation 3.1 and Taylor expanding about $\bar{x}(t)$ gives

$$\dot{x}(t) = \dot{\bar{x}}(t) + \dot{y} = f(\bar{x}(t) + Df(\bar{x}(t))y + \mathcal{O}(|y|^2)) \quad (3.3)$$

where Df is the derivative of f and $|\cdot|$ denotes a norm on $\Re^n$. Using the fact that $\dot{\bar{x}}(t) = f(\bar{x}(t))$, Equation 3.3 becomes

$$\dot{y} = Df(\bar{x}(t))y + \mathcal{O}(|y|^2). \quad (3.4)$$

Equation 3.4 describes the evolution of orbits near $\bar{x}(t)$. For stability questions, we are concerned with the behavior of solutions arbitrarily close to $\bar{x}(t)$, so it seems reasonable that this question could be answered by studying the associated *linear system*:

$$\dot{y}(t) = Df(\bar{x}(t))y. \quad (3.5)$$

Therefore, the question of stability of $\bar{x}(t)$ involves the following two steps:

1. Determine if the $y = 0$ solution of Equation 3.5 is stable.
2. Show that stability (or instability) of the $y = 0$ solution of Equation 3.5 implies stability (or instability) of $\bar{x}(t)$ of the system (Eq. 3.1).

It can be shown that if the eigenvalues of the associated linear vector field have nonzero real parts, then the orbit (i.e., trajectory) structure *near an equilibrium solution* of the nonlinear vector field (Eq. 3.1) is essentially the same as that of the linear vector field (Eq. 3.5). This is addressed by the following theorem.

**Theorem 3.1: Hartman–Grobman
(Hirsch and S. Smale, 1974)**

Consider the system (Eq. 3.1) with an equilibrium point $\bar{x}$. If $Df(\bar{x})$ has no zero or purely imaginary eigenvalues, there is a homeomorphism h defined on a neighborhood U of $\bar{x}$ taking orbits of the flow ϕ_t to those of the linear flow $e^{Df(\bar{x})t}$ of Equation 3.5. The homeomorphism preserves the sense of the orbits and is chosen to preserve parameterization by time.

A corollary of Theorem 3.1 and a fundamental linear system theory lead to the following sufficient condition for an equilibrium point to be asymptotically stable.

**Theorem 3.2: Asymptotic Stability
(Hirsch and Smale, 1974; Sastry, 1999)**

Suppose all of the eigenvalues of $Df(\bar{x})$ of the linear system (Eq. 3.5) have negative real parts. Then, the equilibrium solution $x = (\bar{x})$ of the nonlinear system (Eq. 3.1) is asymptotically stable.

This is a well-known stability evaluation method that checks the eigenvalues of the Jacobian matrix at an equilibrium point. The required computational efforts can be immense. An alternative approach is based on the Lyapunov function theory, which is discussed in the next section.

3.3 LYAPUNOV FUNCTION THEORY

One of the most important developments in the stability theory is the Lyapunov function theory. Lyapunov, a Russian mathematician and engineer, laid down the foundation for the Lyapunov theory. Lyapunov stability theorems give sufficient conditions for Lyapunov stability and asymptotic stability. The appealing feature of the Lyapunov function theory is that it derives the stability properties of the equilibrium point without numerically solving the underlying ordinary differential (difference) equations. This is the “spirit of Lyapunov.” There are several versions of proof of the Lyapunov function theorem.

During the early period of developing direct methods, the Lyapunov function theorem was applied by many researchers to ensure power system transient stability without time-domain simulation. In this section, we present an overview of a fundamental Lyapunov function theorem.

We denote the following notation as the time derivative of a function $V(x)$, taken along the system trajectory:

$$\begin{aligned}
\dot{V}(x(t)) &= \frac{\partial V(x(t))^T}{\partial x} \cdot \dot{x}(t) \\
&= \frac{\partial V(x)^T}{\partial x} \cdot f(x).
\end{aligned} \tag{3.6}$$

Since the vector field $f(x)$ and the gradient of the function $V(x)$ are available without the explicit knowledge of the system trajectory, the time derivative of $V(x(t))$ can be performed without knowledge of the system trajectory.

Theorem 3.3: Lyapunov's Stability (Guckenheimer and Holmes, 1983; Hirsch and Smale, 1974)

Let $\hat{x}$ be an equilibrium point of $\dot{x} = f(x)$, where $f: R^n \rightarrow R^n$. Let $V: U \rightarrow R$ be a continuous function defined on a neighborhood of $\hat{x}$, differentiable on $U \rightarrow \hat{x}$, such that

(a) $V(\hat{x}) = 0$ and $V(x) > 0$ if $x \neq \hat{x}$, and $x \in U$.

(b) $\dot{V}(x) \leq 0$ in $U - \hat{x}$.

Then, $\hat{x}$ is stable. Furthermore, if also

(c) $\dot{V}(x) < 0$ in $U - \hat{x}$, then $\hat{x}$ is asymptotically stable.

(d) $\dot{V}(x) < 0$ in $U - \hat{x}$, $U = R^n$, then $\hat{x}$ is asymptotically stable in the large.

Remarks:

1. The Lyapunov function theory asserts not only the stability property of the equilibrium point (a local result) but also that there does not exist any limit cycle (oscillation behavior) or bounded complicated behavior such as an almost periodic trajectory, chaotic motion, and so on, in the subset of the state space where there exists a Lyapunov function.
2. It should be pointed out that the Lyapunov function theory only furnishes sufficient conditions. If for a particular Lyapunov function candidate, V , the required conditions on the derivative of V , that is, $\dot{V}$, are not met, then conclusions regarding the stability or instability of the equilibrium point still cannot be drawn.

There is no systematic way of constructing Lyapunov functions for general nonlinear systems. This is a fundamental drawback of the Lyapunov direct method. Therefore, faced with specific nonlinear dynamical systems, one often has to use experience, intuition, trial and error, and physical insights (e.g., the energy function for electrical and mechanical systems) to search for an appropriate Lyapunov function. In this literature, a number of methods and techniques facilitating the search of Lyapunov functions have been proposed (Khalil, 2002; Michel et al., 1984; Vaahedi et al., 1998; Vidyasagar, 2002).

3.4 STABLE AND UNSTABLE MANIFOLDS

The concepts of an *invariant set*, a *limit set* including an α -*limit set* and an ω -*limit set*, and *stable* and *unstable manifolds* are important to the dynamical system theory. Each of these concepts is defined next. A detailed discussion of these concepts and implications may be found in Guckenheimer and Holmes (1983), in Paganini and Lesieutre (1999), and in Palis and de Melo (1981).

A set $M \in R^n$ is called an *invariant set* of Equation 3.1 if every trajectory of Equation 3.1 starting in M remains in M for all t . A point p is said to be in the ω -limit set of x if, corresponding to each $\varepsilon > 0$ and $T > 0$, there is a $t > T$ with the property that $|\phi(x, t) - p| < \varepsilon$. This is equivalent to saying that there is a sequence $\{t_i\}$ in R , $t_i \rightarrow \infty$, with the property that $p = \lim_{i \rightarrow \infty} \phi(x, t_i)$. A point p is said to be in the α -limit set of x if, corresponding to each $\varepsilon > 0$ and $T < 0$, there is a $t < T$ with the property that $|\phi(x, t) - p| < \varepsilon$. This is equivalent to saying that there is a sequence $\{t_i\}$ in R , $t_i \rightarrow -\infty$, with the property that $p = \lim_{i \rightarrow \infty} \phi(x, t_i)$. Hence, the ω -limit set captures the asymptotic behaviors of a bounded trajectory in positive time, while the α -limit set captures the asymptotic behaviors of a bounded trajectory in negative time.

One of the fundamental properties of the limit set is as follows.

Theorem 3.4: Properties of Limit Sets

If a trajectory $\phi(x, t)$ of a system (Eq. 3.1) is bounded for $t \geq 0$ (or $t \leq 0$), then its ω -limit set (or α -limit set) exists; moreover, its limit set is compact, connected, and invariant.

Generally speaking, limit sets can be very complex; they can be equilibrium points, limit cycles (closed orbits), quasi-periodic solutions, and chaos. Stable limit sets are of supreme importance in experimental and numerical settings because they are the only kind of limit sets that can be observed naturally! The concept of stable limit sets is similar to that of equilibrium points.

Definition: Stable Limit Set

A limit set L is said to be *Lyapunov stable* if, for each open neighborhood U of L , there exists an open neighborhood V of L , such that for all $x \in V$ and for all $t > 0$, $\phi(x) \in U$. Equivalently, $\phi_t(x) \in U$ for all $t > 0$. Otherwise, L is unstable.

Definition: Asymptotically Stable Limit Set

A limit set L is *asymptotically stable* if there exists an open neighborhood V of L such that the ω -limit set of every point in V is L . Equivalently,

- (i) $\phi_t(x) \in V$, $t > 0$ and
- (ii) $\lim_{t \rightarrow \infty} \inf_{y \in L} \|\phi_t(x) - y\| = 0$.

We next review the concept of stable and unstable manifolds of limit sets. We start from the simplest limit set—the equilibrium point. Let $\hat{x}$ be an equilibrium point

and $U \subset R^n$ be a neighborhood of $\hat{x}$. We define the local stable manifold of $\hat{x}$ as follows:

$$W_{loc}^S(\hat{x}) := \{x \in U : \phi_t(x) \rightarrow \hat{x} \text{ as } t \rightarrow \infty\}.$$

The local unstable manifold of $\hat{x}$ is defined as

$$W_{loc}^u(\hat{x}) := \{x \in U : \phi_t(x) \rightarrow \hat{x} \text{ as } t \rightarrow -\infty\}.$$

Note that $W_{loc}^S(\hat{x})$ is a positive invariant set, while $W_{loc}^u(\hat{x})$ is a negative invariant set. They may not be manifolds when $\hat{x}$ is nonhyperbolic. We next present a fundamental theorem of stable and unstable manifolds for a hyperbolic equilibrium point. Recall that an equilibrium point is *hyperbolic* if the corresponding Jacobian matrix has no eigenvalues with zero real parts; otherwise, it is a nonhyperbolic *equilibrium point*.

Theorem 3.5: Unstable–Stable Manifold Theorem (Guckenheimer and Holmes, 1983; Sastry, 1999)

Suppose that the nonlinear system (Eq. 3.1) has a hyperbolic equilibrium point $\bar{x}$. Then the sets $W_{loc}^S(\bar{x})$ and $W_{loc}^u(\bar{x})$, often referred to as the local stable and unstable manifolds, are manifolds of the same dimensions n_s, n_u as those of the stable and unstable eigenspaces E^s, E^u of the linearized system (Eq. 3.5). These manifolds are also tangent to E^s, E^u at $\bar{x}$. $W_{loc}^S(\bar{x})$ and $W_{loc}^u(\bar{x})$ are as smooth as the vector field $f(x)$ of Equation 3.1 (see Figure 3.3).

The stable manifold $W^S(\bar{x})$ and the unstable manifold $W^u(\bar{x})$ are obtained by letting points in $W_{loc}^S(\bar{x})$ flow backward in time and points in $W_{loc}^u(\bar{x})$ flow forward in time:

$$W^S(\bar{x}) := \bigcup_{t \leq 0} \phi_t(W_{loc}^S(\bar{x})) \text{ and} \quad (3.7)$$

$$W^u(\bar{x}) := \bigcup_{t \geq 0} \phi_t(W_{loc}^u(\bar{x})). \quad (3.8)$$

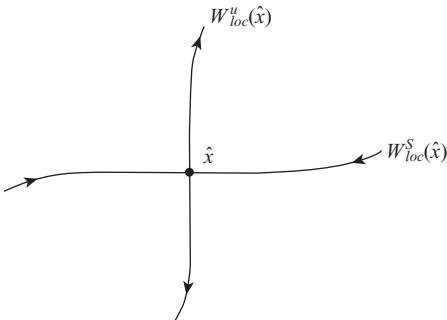


Figure 3.3 The local stable and unstable manifolds of an equilibrium point.

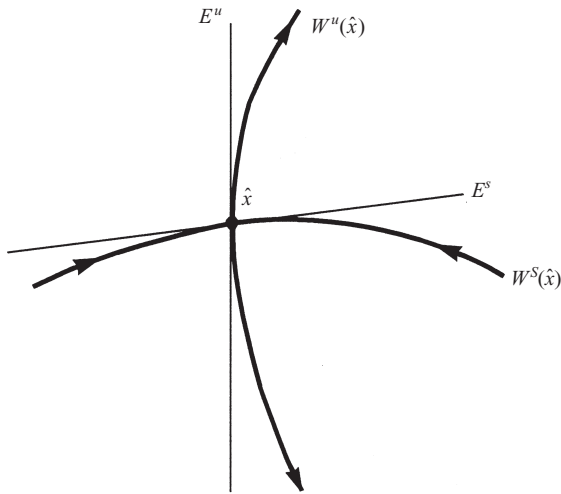


Figure 3.4 The relationship between a stable eigenspace, an unstable eigenspace, and stable and unstable manifolds at a hyperbolic equilibrium point.

Remarks:

1. Clearly, $\hat{x}$ is the ω -limit set of every point in $W^s(\hat{x})$, as well as the α -limit set of every point in $W^u(\hat{x})$. For a hyperbolic equilibrium point, the dimension of the stable manifold $W^s(\hat{x})$ equals the number of eigenvalues of the Jacobian $J_f(\hat{x})$ with negative real parts. The sum of the dimension of $W^s(\hat{x})$ and that of $W^u(\hat{x})$ equals the dimension of the state space (see Figure 3.4).
2. Stable and unstable manifolds are invariant sets. Every trajectory in the stable manifold $W^s(\hat{x})$ converges to $\hat{x}$ as time goes to positive infinity, while every trajectory in the stable manifold $W^u(\hat{x})$ converges to $\hat{x}$ as time goes to negative infinity.
3. The above concepts and definitions of stable and unstable manifolds are applicable to other classes of hyperbolic limit sets such as periodic solutions, quasi-periodic solutions and chaos.

Motivated by Theorem 3.5, we define the type of equilibrium point based on the number of eigenvalues of the corresponding Jacobian matrix with a positive real part.

Definition: Type of Equilibrium Point

The *type* of hyperbolic equilibrium point p of a general nonlinear system (Eq. 3.1) is defined as the number of eigenvalues of $(\partial f/\partial x)(p)$ with a positive real part. If $(\partial f/\partial x)(p)$ has exactly one eigenvalue with a positive real part, we call p a *type-one equilibrium point*. Similarly, p is called a *type- k equilibrium point* if $(\partial f/\partial x)(p)$ has exactly a k eigenvalue with a positive real part.

Type-one equilibrium points play a dominating role in the characterization of a stability boundary and a quasi-stability boundary. Hence, type-one equilibrium points and stable equilibrium points (SEPs) are important equilibrium points used in this text.

The idea of transversality is basic in the study of dynamical systems (Palis, 1969; Palis and de Melo, 1981; Smale, 1967). If A and B are injectively immersed manifolds in M , we say they satisfy the *transversality condition* if either (1) at every point of intersection $x \in A \cap B$, the tangent spaces of A and B span the tangent space of M at x ; that is,

$$T_x(A) + T_x(B) = T_x(M) \quad \text{for } x \in A \cap B,$$

or (2) they do not intersect at all. One of the most important features of a hyperbolic equilibrium point $\hat{x}$ is that its stable and unstable manifolds intersect transversely at $\hat{x}$. This transverse intersection is important because it persists under perturbation of the vector field.

3.5 STABILITY REGIONS

For a SEP, say, x_s , there exists a number $\delta > 0$ such that every point in the set, $\|x_0 - x_s\| < \delta$ implies that the trajectory starting from the initial point x_0 converges to the SEP x_s ; that is, $\phi_t(x_0) \rightarrow \hat{x}$ as $(t \rightarrow \infty)$. If δ is arbitrarily large, then $\hat{x}$ is called a *global SEP*. There are many physical systems containing SEPs but not globally stable equilibrium points. A useful concept for these kinds of systems is that of the *stability region* (also called the *region of attraction*). The stability region of a SEP x_s is the set of all points x such that

$$\lim_{t \rightarrow \infty} \phi_t(x) = x_s. \quad (3.9)$$

We will denote the stability region of x_s by $A(x_s)$ and its closure by $\bar{A}(x_s)$, respectively; hence,

$$A(x_s) := \{x \in R^n : \lim_{t \rightarrow \infty} \phi_t(x) = x_s\}. \quad (3.10)$$

When it is clear from the context, we write A for $A(x_s)$, and so on. Alternatively, the stability region can be expressed as

$$A(x_s) = \{x \in R^n : \omega(x) = x_s\}, \quad (3.11)$$

where $\omega(x)$ denotes the ω -limit set of x . From a topological point of view, the stability region $A(x_s)$ is an open, invariant, and connected set (Hirsch, 1976). The boundary of the stability region $A(x_s)$ is called the *stability boundary* (also called *separatrix*) of x_s and will be denoted by $\partial A(x_s)$ (see Figure 3.5).

We next discuss some topological properties of the stability region and stability boundary. Since the stability region of a SEP is, in fact, the stable manifold, the following topological properties of stability regions are based on the properties of the stable manifold of x_s (Hirsch and Pugh, 1970).

Theorem 3.6: Topological Property

The stability region $A(x_s)$ is an open, invariant set that is diffeomorphic to R^n .

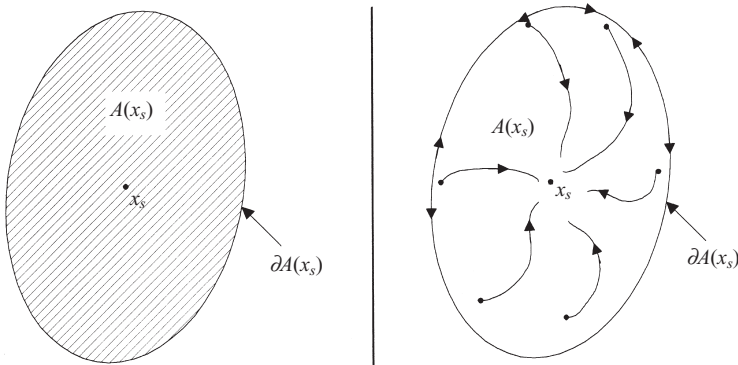


Figure 3.5 As time increases, every trajectory in the stability region $A(x_s)$ converges to the SEP, x_s , and every trajectory on the stability boundary evolve on the boundary.

The above result asserts that every trajectory in a stability region lies entirely in the stability region and that the dimension of the stability region is n . Since the boundary of an invariant set is also invariant and the boundary of an open set is a close set, the set of the stability boundary has the following property.

Theorem 3.7: Topological Property

The stability boundary $\partial A(x_s)$ is a closed invariant set of dimension $< n$. If $A(x_s)$ is not dense in R^n , then $\partial A(x_s)$ is of dimension $n - 1$.

If there are at least two SEPs, then the dimension of each stability boundary is $n - 1$; in particular, stability boundaries are nonempty in this case. Characterization of stability regions can be achieved via characterization of the stability boundary. We will develop a comprehensive theory for stability regions of nonlinear dynamical systems (Eq. 3.1) in the next section.

3.6 LOCAL CHARACTERIZATIONS OF STABILITY BOUNDARY

Our aim is to present a comprehensive theory of stability regions for the (autonomous) dynamical system (Eq. 3.1). In this section, several dynamic and topological properties of a stability boundary will be also derived. A complete characterization for two fundamental limit sets of general nonlinear dynamical systems (i.e., equilibrium points and limit cycles) lying on the stability boundary will be derived.

Our approach starts from a local characterization of the stability boundary and moves towards a global characterization of the stability boundary. We first derive a complete characterization for an equilibrium point to lie on the stability boundary, which is a key step in the characterization of the stability region $A(x_s)$. We do this in two steps. First, we impose only one assumption on the dynamical system, namely, that equilibrium points are hyperbolic, and derive conditions for an equi-

librium point to be on the stability boundary in terms of both its stable and unstable manifolds. Additional conditions are then imposed on the dynamical system and the results are further sharpened. We also derive the characterizations of closed orbits on the stability boundary. We use the notation $A - B$ to denote those elements that belong to A but not to B . We use the term critical element to denote equilibrium points and limit cycles.

Let x be a hyperbolic critical element. Let U be a neighborhood of x in $W^s(x)$ whose boundary ∂U is transversal to the vector field f . We call ∂U a *fundamental domain* of $W^s(x)$. A cross section $V \subset R^n$ of a vector field f is a manifold V of dimension $n - 1$, which need not be a hyperplane but must be in a manner such that the flow of f is everywhere transversal to it. Any cross section of f containing ∂U and transversal to $W^s(x)$ is the so-called *fundamental neighborhood* $G(x)$ associated with $W^s(x)$. It follows that $W^s(x) = \bigcup_{t \in R} \Phi_t(\partial U) \cup \{x\}$ and $\bigcup_{t \geq 0} \Phi_t(G(x)) \cup W^u(x)$ contains a neighborhood of x .

Theorem 3.8: Characterization of Equilibrium Point on the Stability Boundary (Chiang et al., 1988)

Let A be the stability region of a SEP x_s . Let $x \neq x_s$ be a hyperbolic equilibrium point. Then,

- (i) if $\{W^u(\hat{x}) - \hat{x}\} \cap \bar{A} \neq \emptyset$, then $\hat{x} \in \partial A$. Conversely, if $\hat{x} \in \partial A$, then $\{W^u(\hat{x})\} \cap \bar{A} \neq \emptyset$;
- (ii) Suppose $\hat{x}$ is not a source (i.e., $\{W^s(\hat{x}) - \hat{x}\} \neq \emptyset$). Then $\hat{x} \in \partial A$ if and only if $\{W^s(\hat{x}) - \hat{x}\} \cap \partial \bar{A} \neq \emptyset$.

Proof: (i) If $y \in W^u(x) \cap \bar{A}$, then

$$\lim_{t \rightarrow \infty} \Phi_{-t}(y) = \hat{x}.$$

But since $\bar{A}$ is invariant, we have

$$\Phi_{-t}(y) \in \bar{A}.$$

It follows that

$$\hat{x} \in \bar{A}.$$

Since $\hat{x}$ cannot be in the stability region, $\hat{x}$ is on the stability boundary.

Suppose, conversely, that $\hat{x} \in \partial A$. Let $G \subset \{W^u(\hat{x}) - \hat{x}\}$ be a fundamental domain of $W^u(\hat{x})$; this means that G is a compact set, such that

$$\bigcup_{t \in R} \Phi_t(G) = \{W^u(\hat{x}) - \hat{x}\}.$$

Let G_ϵ be the ϵ -neighborhood of G in R^n . Then $\bigcup_{t < 0} \Phi_t(G_\epsilon)$ contains a set of the form $\{U - W^s(\hat{x})\}$, where U is a neighborhood of $\hat{x}$. Since $\hat{x} \in \partial A$, it follows that $U \cap A \neq \emptyset$. But, by assumption, $\hat{x} \in \partial A$, so $W^s(\hat{x}) \cap A = \emptyset$. Therefore, we have

$$\{U - W^s(\hat{x})\} \cap A \neq \emptyset$$

or

$$\bigcup_{t < 0} \Phi_t(G_\varepsilon) \cap A \neq \emptyset.$$

This implies that $G_\varepsilon \cap \Phi_t(A) \neq \emptyset$ for some t . Since A is invariant under the flow, it follows that

$$G_\varepsilon \cap A \neq \emptyset.$$

Since $\varepsilon > 0$ is arbitrary and G is a compact set, we conclude that G contains at least a point of $\bar{A}$. The proof of (ii) is similar to the proof of (i), thus completing the proof.

The above characterization of an equilibrium point lying on the stability boundary can be extended to another critical element, that is, closed orbit (limit cycle). By a *closed orbit* of a dynamical system we mean the image of a nonconstant periodic solution of Equation 3.1; that is, a trajectory γ is a closed orbit if γ is not an equilibrium point and $\Phi_t(x) = x$ for some $x \in \gamma$, $t \neq 0$. A closed orbit γ is said to be *hyperbolic* if, for any $p \in \gamma$, $n - 1$ of the eigenvalues of the Jacobian of $\Phi_t(\gamma)$ at p have a modulus not equal to 1 (one eigenvalue must always be 1). A *critical element* of the vector field f is either a closed orbit or an equilibrium point.

The stable and unstable manifolds of a hyperbolic closed orbit γ are defined as the following:

$$W^s(\gamma) = \{x \in M : \Phi_t(x) \rightarrow \gamma \text{ as } t \rightarrow \infty\} \text{ and}$$

$$W^u(\gamma) = \{x \in M : \Phi_t(x) \rightarrow \gamma \text{ as } t \rightarrow -\infty\}.$$

A characterization of the closed orbit (limit cycle) on the stability boundary is as follows.

Theorem 3.9: Characterization of a Closed Orbit on the Stability Boundary (Chiang et al., 1988)

Let A be the stability region of a SEP. Let γ be a hyperbolic closed orbit. Then,

- (i) $\gamma \subseteq \partial A$ if and only if $\{W^u(\gamma) - \gamma\} \cap \bar{A} \neq \emptyset$;
- (ii) Suppose $\{W^s(\gamma) - \gamma\} \neq \emptyset$. Then, $\gamma \subseteq \partial A$ if and only if $\{W^s(\gamma) - \gamma\} \cap \partial A \neq \emptyset$.

As a corollary to Theorem 3.8, if $\{W^u(\hat{x}) - \hat{x}\} \cap A \neq \emptyset$, then $\hat{x}$ must be on the stability boundary. Since any trajectory in $A(x_s)$ approaches x_s , we see that a sufficient condition for $\hat{x}$ to be on the stability boundary is the existence of a trajectory in $W^u(\hat{x})$, which approaches x_s . The nice thing about this condition is that it can be checked numerically. From a practical point of view, we would like to see when this condition is necessary. We will show that this condition becomes necessary under two additional assumptions.

So far, we have assumed only that the critical elements are hyperbolic. This is a generic property for dynamical systems. Roughly speaking, we say a property is generic for a class of systems if that property is true for *almost all* systems in the class. A formal definition is given in Hirsch (1976). It has been shown in Palis (1969) that among C^r ($r \geq 1$) vector fields, the following properties are generic: (1) all equilibrium points and closed orbits are hyperbolic, and (2) the intersections of the stable and unstable manifolds of critical elements satisfy the transversality condition. Theorem 3.8 can be sharpened under two conditions, one of which is generic for a nonlinear dynamical system (Eq. 3.1). This is the transversality condition. The other condition requires that every trajectory on the stability boundary approaches one of the critical elements.

The following proposition (Hirsch, 1976) is used in the proofs of the next two theorems.

Lemma 3.10

Let x_i and x_j be hyperbolic critical elements of the nonlinear dynamical system (Eq. 3.1). Suppose that the intersection of the stable and unstable manifolds of x_i, x_j satisfy the transversality condition and $\{W^u(x_i) - x_i\} \cap \{W^s(x_j) - x_j\} \neq \emptyset$. Then, $\dim W^u(x_i) \geq \dim W^u(x_j)$, where the equality sign is true only when x_i is an equilibrium point and x_j is a closed orbit.

The following Lemma, which is a weak version of the λ -lemma (Chiang et al., 1988), is useful in the proof of the next theorem. Recall that the type of an equilibrium point is the dimension of its unstable manifold. An m -disk is a disk of dimension m .

Lemma 3.11

Let $\hat{v}$ be a hyperbolic critical element of the nonlinear dynamical system (Eq. 3.1) with dimension $W^u(\hat{v}) = m$. If $\hat{v}$ is an equilibrium point, let D be an m -disk in $W^u(\hat{v})$. If $\hat{v}$ is a closed orbit, let D be an $(m - 1)$ -disk in $W^u(\hat{v}) \cap S$, where S is a cross section at $p \in \hat{v}$. Let N be an m -disk (if $\hat{v}$ is an equilibrium point) or $(m - 1)$ -disk (if $\hat{v}$ is a closed orbit) having a point of transversal intersection with $W^s(\hat{v})$. Then D is contained in the closure of the set $\cap_{t \geq 0} \Phi_t(N)$.

Now, we present the key theorem of this section, which characterizes an equilibrium point being on the stability boundary in terms of both its stable and unstable manifolds. From the practical point of view, this result is more useful in the numerical verification of equilibrium points on the stability boundary than Theorem 3.8.

Theorem 3.12: Characterization of an Equilibrium Point on the Stability Boundary (Chiang et al., 1988)

Let A be the stability region of a SEP of the nonlinear dynamical system (Eq. 3.1). Let $\hat{x}$ be an equilibrium point. Assume the following.

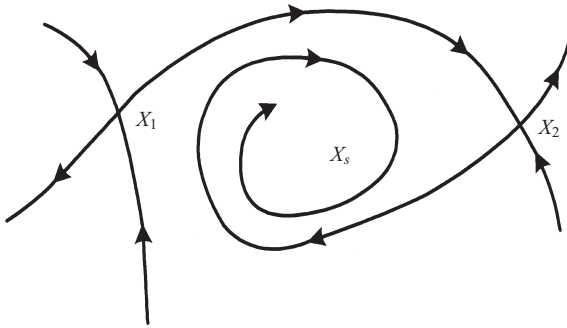


Figure 3.6 The intersection between the unstable manifold of x_1 and the stable manifold of x_2 does not satisfy the transversality condition.

- (A1) All the equilibrium points on ∂A are hyperbolic.
- (A2) The stable and unstable manifolds of equilibrium points on ∂A satisfy the transversality condition.
- (A3) Every trajectory on ∂A approaches one of the equilibrium points as $t \rightarrow \infty$.

Then,

- (1) $\hat{x} \in \partial A$ if and only if $W^u(\hat{x}) \cap A \neq \emptyset$ and
- (2) $\hat{x} \in \partial A$ if and only if $W^s(\hat{x}) \subseteq \partial A$.

To show that the transversality condition is needed in Theorem 3.12, let us consider the example taken from Tsolas et al. (1985). In Figure 3.6, the transversality condition is not satisfied because the intersection of the unstable manifold of x_1 and the stable manifold of x_2 is a portion of the manifold whose tangent space has a dimension of 1. Note that the unstable manifold of x_1 intersects with the stability boundary (see Theorem 3.8) but not the stability region (see Theorem 3.12). A part of the stable manifold of x_1 (upper part in Figure 3.6) is not in the stability boundary (see Theorem 3.12).

Theorem 3.13 below extends the result of Theorem 3.12 to accommodate closed orbits on the stability boundary.

Theorem 3.13: Characterization of a Critical Element on the Stability Boundary (Chiang et al., 1988)

Let A be the stability region of a SEP of the nonlinear dynamical system (Eq. 3.1). Let r be a critical element. Assume the following.

- (B1) All the critical elements on ∂A are hyperbolic.
- (B2) The stable and unstable manifolds of critical elements on ∂A satisfy the transversality condition.
- (B3) Every trajectory on ∂A approaches one of the critical elements as $t \rightarrow \infty$.

Then,

- (1) $\hat{r}$ is on the stability boundary ∂A if and only if $W^u(\hat{r}) \cap A \neq \emptyset$ and
- (2) $\hat{r}$ is on the stability boundary ∂A if and only if $W^s(\hat{r}) \subseteq \partial A$.

The next result concerns the number of equilibrium points on the stability boundary. We say that $S \subset R^n$ is a *smooth* manifold of dimension s if, for each point $p \in S$, there exist a neighborhood $U \subset S$ of p and a homeomorphism $h: U \rightarrow V$, where V is an open subset of R^s , such that the inverse homeomorphism $h^{-1}: V \rightarrow U \subset R^n$ is an immersion of class C^1 .

Theorem 3.14: Number of Equilibrium Points on the Stability Boundary

If the stability boundary ∂A of a SEP is a smooth compact manifold and all the equilibrium points on ∂A are hyperbolic, then the number of equilibrium points on ∂A is even.

Proof: The proof is based on the following fact (Hirsch, 1976, p. 139, Exercise 7): the Euler characteristic of the boundary of a compact manifold is even. From the Poincaré–Hopf index theorem (Guillemin and Pollack, 1974, p. 134), it follows that the sum of the indices of equilibrium points of f on the smooth, compact stability boundary ∂A are even. But the index of f at a hyperbolic equilibrium point is either $+1$ or -1 (Milnor, 1965, p. 37). Consequently, Theorem 3.14 follows.

3.7 GLOBAL CHARACTERIZATION OF STABILITY BOUNDARY

In this section, we characterize the stability boundary for a fairly large class of the nonlinear dynamical system (Eq. 3.1) whose stability boundary is nonempty. We make the following assumptions concerning the vector field:

- (A1) All the equilibrium points on the stability boundary are hyperbolic.
- (A2) The stable and unstable manifolds of equilibrium points on the stability boundary satisfy the transversality condition.
- (A3) Every trajectory on the stability boundary approaches one of the equilibrium points as $t \rightarrow \infty$.

Remark: Assumption (A1) is a generic property of C^1 dynamical systems and can be checked for a particular system by direct computation of the eigenvalues of the corresponding Jacobian matrix of the vector field. Assumption (A2) is also a generic property; however, it is not easy to check. Assumption (A3) is not a generic property, but in many systems, it can be verified by several means.

Theorem 3.15 asserts that if Assumptions (A1)–(A3) are satisfied, then the stability boundary is the union of the stable manifolds of the equilibrium points on the stability boundary (Zaborszky et al., 1988a).

Theorem 3.15: Characterization of the Stability Boundary

For a nonlinear autonomous dynamical system (Eq. 3.1) which satisfies Assumptions (A1)–(A3), let x_i , $i = 1, 2, \dots$ be the equilibrium points on the stability boundary ∂A of the SEP. Then,

- [1] $x_i \in \partial A$ if and only if $W^u(x_i) \cap A \neq \emptyset$ and
- [2] $\partial A = \bigcup_i W^s(x_i)$.

Proof: Part [1] is shown in Theorem 3.12. We prove part [2]. Let x_i , $i = 1, 2, \dots$ be the equilibrium points on the stability boundary ∂A . Theorem 3.12 implies the following:

$$\partial A \supseteq \bigcup_i W^s(x_i). \quad (3.12)$$

Assumption (A3) implies

$$\partial A \subseteq \bigcup_i W^s(x_i). \quad (3.13)$$

Combining Equations 3.12 and 3.13, we complete the proof for part [2].

Theorem 3.15 can be generalized to allow closed orbits to exist on the stability boundary. The following theorem gives an interesting result on the structure of the equilibrium points on the stability boundary. Moreover, it presents a necessary condition for the existence of certain types of equilibrium points on a *bounded* stability boundary.

Theorem 3.16: Structure of Equilibrium Points on the Stability Boundary

For the nonlinear autonomous dynamical system (3.1) containing two or more SEPs, if it satisfies Assumptions (A1)–(A3), then the stability boundary must contain at least one type-one equilibrium point. If, furthermore, the stability region is bounded, then ∂A must contain at least one type-one equilibrium point and one source.

Proof: Since there are at least two SEPs including, say, x_s , it follows that the dimension of $\partial A(x_s)$ is $(n - 1)$. Since $\partial A(x_s) = \bigcup W^s(x_j)$, where $x_j \in \partial A(x_s)$, at least one of the x_j must be a type-one equilibrium point, say, x_1 , so that the dimensions of $\bigcup W^s(x_j)$ is $(n - 1)$. Repeating the same argument, if $\partial W^s(x_1)$ is nonempty, then the dimension of $\partial W^s(x_1)$ is $\leq (n - 2)$, say, $(n - k)$. The application of Theorem 3.15 yields $\partial W^s(x_1) = \bigcup W^s(x_j)$, $x_j \in \partial W^s(x_1)$. In order for $\bigcup W^s(x_j)$ to have dimension $(n - k)$, at least one of the x_j must be a type- k equilibrium point. If the stability region is bounded, the same argument can be repeated until we reach a type- n equilibrium point (a source). This completes the proof.

The contrapositive of Theorem 3.16 leads to the following corollary, which is useful in predicting the unboundedness of the stability region.

Theorem 3.17: Sufficient Condition for an Unbounded Stability Region

Consider the nonlinear autonomous dynamical systems (Eq. 3.1) with a SEP x_s whose stability boundary is nonempty. If Assumptions (A1)–(A3) are satisfied and if $\partial A(x_s)$ contains no source, then the stability region $A(x_s)$ is unbounded.

3.8 ALGORITHM TO DETERMINE THE STABILITY BOUNDARY

Theorem 3.15 leads to the following conceptual algorithm for determining the stability boundary of a SEP of the nonlinear autonomous dynamical systems (Eq. 3.1) that satisfy Assumptions (A1)–(A3).

Algorithm: To determine the stability boundary $\partial A(x_s)$,

- Step 1.** Find all the equilibrium points.
- Step 2.** Identify those equilibrium points whose unstable manifolds contain trajectories approaching the SEP x_s .
- Step 3.** The stability boundary of x_s is the union of the stable manifolds of the equilibrium points identified in Step 2.

Step 1 in the algorithm involves finding all the solutions of $f(x) = 0$. Step 2 can be accomplished numerically. The following procedures are suggested:

1. Find the Jacobian at the equilibrium point (say, $\hat{x}$).
2. Find many of the generalized unstable eigenvectors of the Jacobian having unit length.
3. Find the intersection of each of these normalized, generalized, and unstable eigenvectors (say, y_i) with the boundary of an ε -ball of the equilibrium point (the intersection points are $\hat{x} + \varepsilon y_i$ and $\hat{x} - \varepsilon y_i$).
4. Integrate the vector field backward (reverse time) from each of these intersection points up to some specified time. If the trajectory remains inside this ε -ball, then go to next step. Otherwise, we replace the value ε by $\alpha \varepsilon$ and also the intersection points $\hat{x} \pm \varepsilon y_i$ by $\hat{x} \pm \alpha \varepsilon y_i$ where $0 < \alpha < 1$. Repeat this step.
5. Numerically integrate the vector field starting from these intersection points.
6. Repeat steps 3–5. If any of these trajectories approach x_s , then the equilibrium point is on the stability boundary.

For a planar system, the equilibrium point on the stability boundary is either a type-one equilibrium point or a type-two equilibrium point, which is a source. The stable manifold of a type-one equilibrium point in this case has dimension one, which can easily be determined numerically as follows:

1. Find a normalized stable eigenvector y of the Jacobian at the equilibrium point $\hat{x}$.
2. Find the intersection of this stable eigenvector with the boundary of an ε -ball of the equilibrium point $\hat{x}$ (where the intersection points are $\hat{x} + \varepsilon y_i$ and $\hat{x} - \varepsilon y_i$).
3. Integrate the vector field from each of these intersection points after some specified time. If the trajectory remains inside this ε -ball, then go to next step. Otherwise, we replace the value ε by $\alpha\varepsilon$ and also the intersection points $\hat{x} \pm \varepsilon y_i$ by $\hat{x} + \alpha\varepsilon y_i$, where $0 < \alpha < 1$. Repeat this step.
4. Numerically integrate the vector field backward (reverse time) starting from these intersection points.
5. The resulting trajectories are the stable manifold of the equilibrium point.

For higher-dimensional systems, the numerical procedure similar to the one above can only provide a set of trajectories on the stable manifold. Finding the stable and unstable manifolds of an equilibrium point is a nontrivial problem, and advanced numerical methods for computing stable and unstable manifolds are needed (Ushiki, 1980).

The method for a complete determination of a stability region will be illustrated on two simple examples. In each example, two figures will be derived: one compares the estimated stability region by the previous methods and the present method, while the other gives the phase portrait of the system to verify the results of this method. Throughout these examples, we *assume* the transversality condition Assumption (A2) is satisfied.

Example 1: This is an example studied in Genesio and Vicino (1984b) and in Michel et al. (1982):

$$\begin{aligned}\dot{x}_1 &= -2x_1 + x_1x_2 \\ \dot{x}_2 &= -x_1 + x_1x_2.\end{aligned}\tag{3.14}$$

There are two equilibrium points: (0.0, 0.0) is a SEP and (1, 2) is a type-one equilibrium point. Assumption (A1) is satisfied. The trajectory on the unstable manifold of (1, 2) converges to the SEP (0.0, 0.0); hence, (1, 2) is on the stability boundary (see Theorem 3.8). Next, we check Assumption (A3). Consider the following function:

$$V(x_1, x_2) = x_1^2 - 2x_1x_2 + x_2^2.$$

The derivative of $V(x_1, x_2)$ along the trajectory of Equation 3.14 is

$$\begin{aligned}\dot{V}(x_1, x_2) &= \frac{\partial V}{\partial x_1}x_1 + \frac{\partial V}{\partial x_2}x_2 \\ &= -2(2x_1 - x_2)(x_1 - x_2).\end{aligned}$$

Hence,

$$\dot{V}(x_1, x_2) < 0 \quad \text{for } (x_1, x_2) \in B^C := R^2 - B,$$

where $B := \{(x_1, x_2) : 2x_1 - x_2 \geq 0 \text{ and } x_1 - x_2 \leq 0\}$.

Define the following sets:

$$\tilde{B} := B_1 \cup B_2 \cup B_3,$$

where

$$B_1 = \{(x_1, x_2) : x_1 < 1, x_2 < 2\},$$

$$B_2 = \{(x_1, x_2) : x_1 \geq 1, x_2 \leq 2\} \cap B, \text{ and}$$

$$B_3 = \{(x_1, x_2) : x_1 > 1, x_2 > 2\}.$$

Since in the set B_1 both $|x_1(t_i)|$ and $|x_2(t_i)|$ are strictly decreasing sequences, we conclude that B_1 is inside the stability region $(0, 0)$. In other words, the stability boundary $\partial A(0, 0)$ cannot lie in B_1 . On the other hand, every trajectory of Equation 3.14 in the set B_3 is unbounded as $t \rightarrow \infty$; therefore, the stability boundary $\partial A(0, 0)$ cannot lie in B_3 either. However, by checking the vector field of Equation 3.14 in B_2 , we find that every trajectory in B_2 will either enter into B_1 or B_3 or converge to the point $(1, 2)$. Hence, we have shown that the stability boundary $\partial A(0, 0)$ cannot be in B_1 nor B_3 ; the part of the stability boundary $\partial A(0, 0)$ in B_2 must converge to $(1, 2)$. Next, we will show that the part of the stability boundary $\partial A(0, 0)$ in $R^2 - \tilde{B}$ also converges to $(1, 2)$. Then, we may claim that Assumption (A3) is satisfied. Note that $\dot{V}(x_1, x_2) < 0$ for $(x_1, x_2) \in R^2 - \tilde{B} \subseteq B^C$ and the function $V(x_1, x_2)$ is a proper map in $R^2 - \tilde{B}$. Thus, every trajectory of $\partial A(0, 0)$ in $R^2 - \tilde{B}$ is bounded and, if it converges in $R^2 - \tilde{B}$, it must converge to an equilibrium point in $R^2 - \tilde{B}$. However, there is no equilibrium point in $R^2 - \tilde{B}$. So, the part of the stability boundary $\partial A(0, 0)$ in $R^2 - \tilde{B}$ must enter the set $\tilde{B}$. However, we have shown that the stability boundary $\partial A(0, 0)$ in $\tilde{B}$ converges to $(1, 2)$. Therefore, the trajectories on the stability boundary $\partial A(0, 0)$ converge to $(1, 2)$, and Assumption (A3) is shown to be satisfied.

It follows that the stability boundary is the stable manifold of $(1, 2)$ (see Theorem 3.15), which is curve C in Figure 3.7. Because there is no source, the stability region is unbounded (Theorem 3.17). Curves A and B in Figure 3.7 are

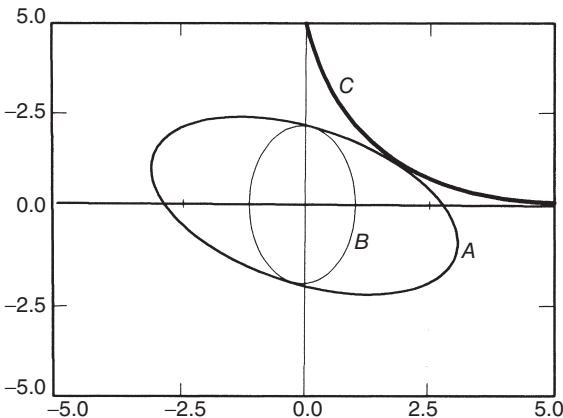


Figure 3.7 Predictions of the stability region of Example 1 by different methods. Curves A and B are obtained by the methods in Michel et al. (1982) and in Genesio and Vicino (1984b). Curve C is obtained by the present method.

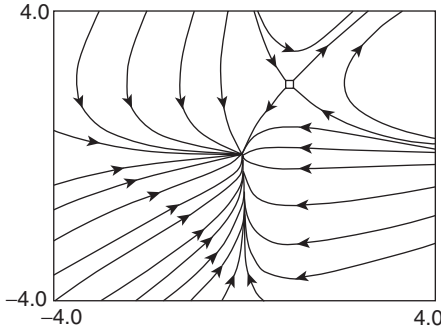


Figure 3.8 The phase portrait of this system. Note that all the points inside curve C converge to the stable equilibrium point, which verified that curve C is the exact stability boundary.

obtained by the methods in Michel et al. (1982) and in Genesio and Vicino (1984b), respectively. The approximately true stability boundary mentioned in Genesio and Vicino (1984b) seems to agree with curve C . Figure 3.8 plots the phase portrait of this system, which confirms that curve C represents the exact stability boundary.

Example 2: The following system is close to a power system transient stability model and is considered in Athay et al. (1979):

$$\dot{x}_2 = 0.301 - \sin(x_1 + 0.4136) + 0.138 \sin 2(x_1 + 0.4136) - 0.279x_2. \quad (3.15)$$

It is straightforward to verify that Assumption (A3) is satisfied with this simple example. The equilibrium points of Equation 3.15 are periodic on the subspace $\{x_1, x_2 | x_2 = 0\}$ and the Jacobian matrix of Equation 3.15 at (x_1, x_2) is

$$J(x) = \begin{bmatrix} 0 & 1 \\ a & -0.279 \end{bmatrix}, \quad (3.16)$$

where $a = -\cos(x_1 + 0.4136) + 0.276 \cos 2(x_1 + 0.4136)$. Let λ_1, λ_2 be the eigenvalue of $J(x)$:

$$\lambda_1 + \lambda_2 = -0.279$$

$$\lambda_1 \times \lambda_2 = -a.$$

The following observations are immediate:

1. Assumption (A1) is satisfied.
2. At least one of the eigenvalues must be negative, which implies there is no source in the system (Eq. 3.15). By Theorem 3.17 we conclude that the stability region (with respect to any SEP) is unbounded.
3. The SEPs and the type-one equilibrium points are located alternately on the x_1 -axis.

It can be shown that $(6.284098, 0.0)$ is a SEP of Equation 3.15. Let us consider its stability region. The application of Theorem 3.8 shows that the type-one equilibrium points $(2.488345, 0.0)$ and $(8.772443, 0.0)$ are on the stability boundary. The

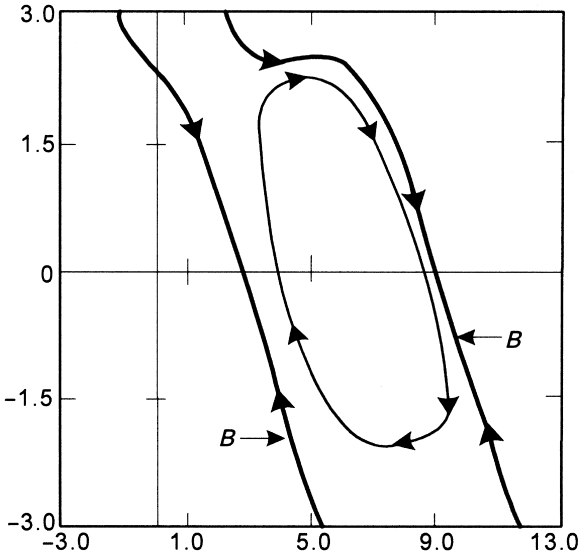


Figure 3.9 Estimations of the stability region of Example 2 by different methods. Curve A is the estimated stability boundary from the methods in Michel et al. (1982) (after a shift in coordinates). Curve B is the estimated stability boundary by the present method.

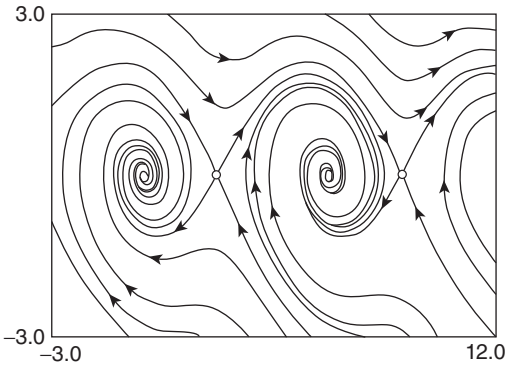


Figure 3.10 The phase portrait of this system, which confirms that curve B in Figure 3.9 is the exact stability boundary.

stability region is again unbounded due to the absence of a source. The stability boundary obtained by the present method is curve B shown in Figure 3.9, which is the union of stable manifolds of the equilibrium points (2.488345, 0.0) and (8.772443, 0.0). Curve A is the estimated stability boundary obtained by the method described in Michel et al. (1982) (after a shift in coordinates). It is clear from the phase portrait in Figure 3.10 that the trajectories of the points inside curve B converge to the SEP, which verifies that curve B is the exact stability boundary.

3.9 CONCLUSION

Depending on the intended use of stability analysis and design, stability can be formulated in a variety of different ways. In this chapter, we have reviewed some

relevant stability concepts from nonlinear dynamical systems theory. One of the most important developments in stability theory is the Lyapunov function theory. An overview of the Lyapunov function theory has been presented. In terms of application, characterization and estimation of the stability region of a SEP are very important. In this chapter, a comprehensive theory of stability regions of SEPs has been presented.

A complete dynamic characterization of the stability boundary of a fairly large class of nonlinear autonomous dynamical systems has been derived. Practical numerical methods are needed to accurately estimate the stability regions of nonlinear dynamical systems and are important to many disciplines in the sciences and engineering (Genesio and Vicino, 1984b; Locufier and Noldus, 2000; Luyckx et al., 2004; Miyagi and Yamashita, 1986; Vittal and Michel, 1986). Presently, few methods are computationally practical and also do not yield accurate results for general large-scale nonlinear dynamical systems are available.

The analytical characterizations of equilibrium points on the stability boundary and the complete characterization of the stability boundary are very useful in the development of theoretical foundations for direct methods of power system transient stability analysis. Furthermore, these analytical results in combination with explorations of the structures of power system transient stability models will lead to the development of practical direct methods for large-scale power systems. These developments will be presented in later chapters.